# Empirical Analysis of Proposer-Builder Separation on Ethereum: Auction Dynamics, MEV Volatility Clustering, and Market Concentration

Kyle D. Rothwell*[1] and Hao Fu†[2]

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**Abstract**

We study Proposer-Builder Separation (PBS) on Ethereum using 1,271,079 delivered payload observations from six MEV-Boost relays (September 2024–May 2026), supplemented by 1,220,100 matched slot-level auction observations from the Relayscan.io Bid Archive [Relayscan, 2024] — the first large-scale empirical characterisation of within-slot bidding behaviour over a 20-month horizon.

Block values are approximately 7.0× higher during high-volatility regimes (Spearman $\rho = 0.494$, $p < 10^{-300}$). GARCH(1,1) reveals extremely persistent volatility ($\hat{\alpha} + \hat{\beta} = 0.980$) with heavy tails ($\hat{\nu} = 4.005$); this persistence is unchanged when isolating the MEV-specific return component, confirming it is not an artifact of ETH price co-movement. Builder concentration (HHI) declined from approximately 2,100 to 600–1,000 over the sample, while both intraday and day-of-week seasonality are highly significant ($p < 10^{-300}$).

The PBS auction is highly competitive at the median: 1,220,100 matched slots show a median of 132 bids from 11 builders per slot, with a median bid margin of 0.141%. Margins compress monotonically with block value (0.196% to 0.110% across terciles; $\rho = -0.054$, 95% CI $[-0.075, -0.031]$) and with relay participation ($\rho = -0.057$). ETH price explains essentially zero variance in block value in ETH terms ($R^2 = 0.0001$), confirming MEV is an on-chain phenomenon. Builder strategies cluster into two genuine types after correcting for a discovered relay-level bid-timestamp artifact: an initially apparent third "post-slot-time" cluster was found to reflect relay-specific data collection characteristics rather than builder behaviour, a finding with implications for cross-relay timing comparisons in MEV-Boost research. These findings carry direct implications for PBS mechanism design, validator economics, and censorship resistance analysis.

**Keywords:** Proposer-Builder Separation, Maximal Extractable Value, Ethereum, GARCH, market microstructure, auction theory, decentralised finance, blockchain economics

*Stevens Institute of Technology, Department of Computer Science. Email: kdrothwell48@icloud.com.

†Stevens Institute of Technology, Department of Financial Engineering.

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# 1 Introduction

Ethereum's transition to proof-of-stake consensus, commonly referred to as the Merge, fundamentally restructured how blocks are produced and how value is extracted from transaction ordering. Prior to the Merge, miners held full control over block construction and could exploit that position to extract Maximal Extractable Value (MEV), which refers to profits arising from the ability to selectively include, exclude, or reorder transactions within a block [Daian et al., 2020]. Following the Merge, Ethereum adopted Proposer-Builder Separation (PBS) via MEV-Boost [Flashbots, 2022], separating the roles of block construction and block proposal into a competitive auction marketplace. Under PBS, validators (proposers) no longer construct blocks themselves. Instead, specialized agents called builders assemble blocks designed to maximize extractable transaction value and submit bids to relay servers. Proposers select the highest-bidding block header and receive a payment from the winning builder. This design is intended to democratize MEV by separating it from the staking process, enabling proposers to capture competitive MEV revenues without requiring sophisticated order flow infrastructure [Buterin, 2021].

Despite PBS's importance to Ethereum's economic security and decentralization goals, several empirical questions remain unresolved. How does MEV opportunity vary with market volatility conditions? Does block value volatility exhibit the temporal clustering observed in traditional financial markets? Has builder competition increased or decreased over time? Are there predictable temporal patterns in MEV activity? And how competitive is the PBS auction itself—how many builders participate, how closely do they compete, and does competition intensify when block values are highest?

We address these questions using 1,271,079 delivered payload observations from six MEV-Boost relays spanning September 2024 through May 2026, supplemented by 1,220,100 slot-level bid observations from the Relayscan.io Bid Archive [Relayscan, 2024]. To our knowledge, this is the first application of GARCH volatility modeling to MEV block values at this scale, and the first large-scale empirical characterization of within-slot bidding dynamics over a 20-month horizon.

The paper is organized as follows. Section 2 reviews related work. Section 3 describes the data. Section 4 presents the methodology. Section 5 reports the findings. Section 6 discusses implications. Section 7 concludes.

# 2 Background and Related Work

MEV was first systematically documented by Daian et al. [2020], who identified transaction-ordering dependencies on Ethereum and estimated that MEV extraction yielded hundreds of millions of dollars annually. Qin et al. [2022] subsequently conducted a comprehensive empirical assessment across MEV categories, finding that arbitrage and liquidations account for the majority of extracted value by magnitude.

PBS was formally proposed by Buterin [2021] as a mechanism to decouple MEV extraction from consensus participation, reducing the centralizing pressure that MEV imposes on the validator set. The practical implementation, MEV-Boost, was developed by Flashbots [2022] and adopted by the majority of Ethereum validators following the Merge. Bahrani et al. [2024] analyzed equilibrium bidding behavior in the PBS auction, establishing theoretical benchmarks for the distribution of surplus between builders and proposers. Wahrstätter et al. [2023] examined builder concentration from a censorship resistance perspective.

Recent empirical work has illuminated the role of CEX-DEX arbitrage in driving high-value block construction. Heimbach et al. [2024] documented a significant correlation between arbitrage

volumes and token price volatility, and showed that builders integrated with CEX-DEX searchers have greater odds of winning MEV-Boost auctions during high-volatility periods. Wu et al. [2025] estimated that 19 major CEX-DEX searchers extracted $233.8M USD over a 19-month period, and demonstrated that exclusive searcher-builder arrangements drive centralization in both markets. These supply-side findings complement the auction-side evidence in the present paper: the 7.0× block value increase in high-volatility regimes documented in Section 5.1 is consistent with CEX-DEX arbitrage flow as the primary mechanism, while the bid margin compression in Section 5.5 reflects the competitive response of builders to that elevated MEV opportunity.

The volatility modeling in this paper draws on GARCH models as introduced by Bollerslev [1986], building on Engle [1982]'s original ARCH framework. The Integrated GARCH specification, in which $\alpha + \beta = 1$, was studied by Engle and Bollerslev [1986] and is associated with permanent effects of volatility shocks.

For market structure analysis, the Herfindahl-Hirschman Index [Hirschman, 1945] and Gini coefficient [Gini, 1912] provide complementary perspectives on concentration and distributional inequality respectively.

# 3 Data

## 3.1 Delivered Payload Collection

Data were assembled by querying the public REST APIs of six MEV-Boost relay operators: Flashbots, Ultra Sound Money, BloXroute (regulated), Aestus, Agnostic Gnosis, and Eden Network. For each slot, only the highest-value delivered payload across all relays was retained to avoid double-counting blocks delivered through multiple relays. Table 1 summarizes the records collected by relay.

Relay selection was governed primarily by data availability rather than a stratified sampling design. These six relays were targeted because they expose (or were expected to expose) a public, unauthenticated `delivered_payloads` endpoint; relays requiring authenticated access were excluded by necessity. Within this set, actual coverage varied substantially due to rate limiting, request timeouts, and differences in API uptime over the collection window, as reflected in the "Coverage" column of Table 1. As a result, the combined dataset is a convenience sample of publicly-queryable relay data rather than a representative cross-section of MEV-Boost market share, and is dominated by BloXroute, which alone contributed approximately 79% of the deduplicated sample (Section 6.5). The empirical findings in Section 5 should accordingly be understood as characterizing the delivered-payload sequence observed through this collection process, not the PBS market as a whole; Section 6.5 discusses the implications of this composition for generalizability.

Table 1: MEV-Boost relay data collection summary

| Relay | Records | Coverage |
|---|---|---|
| Flashbots | 58,700 | Partial (rate limited) |
| Ultra Sound Money | 108,000 | Partial (timeout) |
| BloXroute | 1,000,000 | Sep 2024 – May 2026 |
| Aestus | 16,600 | Partial (timeout) |
| Agnostic | 297,000 | Jul 2024 – Mar 2026 |
| Eden Network | — | Unavailable |
| **Combined (deduplicated)** | **1,271,079** | **Sep 2024 – May 2026** |

### 3.2 Bid Archive Data

To characterise within-slot bidding behaviour, we supplement the delivered payload dataset with bid-level data from the Relayscan.io Bid Archive [Relayscan, 2024], a public archive of MEV-Boost builder bids collected continuously across all major relays since June 2024. For each calendar day, the archive provides two CSV files: one containing all observed bids and one containing only the top bid per builder per slot (the "top bid" file). We download the top-bid files for the period September 2024 through May 2026, yielding 608 daily files of approximately 60–70 MB each (compressed).

For each slot, the bid archive records the slot number, bid value in Wei, the submitting builder's public key, the relay through which the bid was submitted, and timing fields including the timestamp in milliseconds and the elapsed time since slot start (`slot_t_ms`). Bids arrive from three source types: Data API polling (source type 1), the Ultrasound top-bid websocket stream (source type 2), and `GetHeader` polling (source type 0); the majority of observations in the top-bid files are source type 2, which provides the richest timing information.

We match bid-archive observations to delivered payload records by `slot`, retaining for each slot the set of builder bids observed across all relays and source types. Of the 1,271,079 delivered payload slots, 1,220,100 (96.0%) are matched to at least one bid-archive observation, yielding the auction analysis dataset used in Section 5.5. The 4.0% of slots without a bid-archive match arise primarily from six days in the sample for which the downloaded archive file was corrupted; these slots are excluded from auction analysis but retained in all other analyses.

Table 2 summarises the bid archive dataset.

Table 2: Bid archive dataset summary, September 2024 – May 2026

| Metric | Value |
|---|---|
| Calendar days covered | 608 |
| Slots matched to delivered payloads | 1,220,100 |
| Match rate | 96.0% |
| Median bids per slot | 132 |
| Median unique builders per slot | 11 |
| Median unique relays per slot | 3 |

### 3.3 Variable Construction

Block values were converted from Wei to ETH by dividing by $10^{18}$. Timestamps were derived from slot numbers using the Ethereum Beacon Chain genesis timestamp $t_0 = 1{,}606{,}824{,}023$ (December 1, 2020) and the fixed slot duration of 12 seconds:

$$t_{\text{slot}} = t_0 + \text{slot} \times 12 \tag{1}$$

A common error in Ethereum data analysis is to substitute the Merge timestamp for genesis, which displaces all computed dates by approximately 18 months; all analyses in this paper use the correct genesis timestamp. These slot-derived timestamps are used directly in Section 5.4 to assign each delivered payload to an hour-of-day (UTC) and day-of-week for the temporal pattern analysis.

Log returns were computed as:

$$r_t = \log(1 + v_t) - \log(1 + v_{t-1}) \tag{2}$$

where $v_t$ denotes block value in ETH at slot $t$. Realized volatility was estimated using a rolling window of 300 delivered-payload observations:

$$\hat{\sigma}_t = \sqrt{300} \cdot \text{std}(r_{t-299}, \ldots, r_t) \tag{3}$$

Because delivered payloads represent approximately 28% of all Ethereum slots over the sample period, consecutive observations in the dataset are not uniformly spaced in clock time; the 300-observation window covers approximately 3–4 hours of clock time on average rather than one hour.

### 3.4 Descriptive Statistics

The mean block value across the full sample is 0.054 ETH, compared to a median of 0.016 ETH, with a maximum of 1,243.3 ETH. The large ratio of mean to median and the extreme upper tail confirm a heavily right-skewed, fat-tailed distribution in which most blocks carry modest value while rare extreme events dominate aggregate MEV extraction. These distributional features motivate the use of non-parametric statistical tests and the Student's $t$ error distribution in GARCH estimation. All code and data collection scripts are publicly available at `https://github.com/kdrsoccer/pbs-mev-research`.

## 4 Methodology

### 4.1 Volatility Regime Analysis

Observations were partitioned into three volatility regimes based on the 33rd and 67th percentiles of the realized volatility series $\hat{\sigma}_t$ (eq. 3): Low ($\hat{\sigma}_t \leq 0.6554$), Mid ($0.6554 < \hat{\sigma}_t \leq 1.4340$), and High ($\hat{\sigma}_t > 1.4340$), where 0.6554 and 1.4340 are the empirical 33rd and 67th percentiles of $\hat{\sigma}_t$ over the full sample.

Group differences were tested using the Kruskal-Wallis test:

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(n+1) \tag{4}$$

where $R_i$ is the rank sum of group $i$ and $n_i$ is the group size. A directional hypothesis (Low < High) was tested using the Mann-Whitney U test. The monotone association between realized volatility and block value was measured by Spearman's $\rho$.

The bootstrap procedure used to obtain the confidence interval for $\hat{\rho}$ draws $n = 8{,}000$ observations via simple random sampling without replacement from the pooled 1,271,079 observations. Because the underlying series exhibits volatility persistence (Section 5.2), this procedure may understate the true sampling variability of $\hat{\rho}$ relative to a block bootstrap that preserves serial dependence; the reported 95% CI [0.477, 0.510] should accordingly be treated as a lower bound on the true confidence interval width.

### 4.2 GARCH Volatility Modeling

ARCH effects were tested using the Ljung-Box statistic on squared log returns. A GARCH(1,1) model with constant mean and Student's $t$ errors was then estimated:

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad z_t \overset{iid}{\sim} t_\nu \tag{5}$$

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \tag{6}$$

Volatility persistence is measured by $\alpha+\beta$; values close to unity indicate highly persistent volatility dynamics approaching Integrated GARCH behavior. As delivered payloads are not uniformly spaced in clock time, the GARCH model characterizes the volatility dynamics of the delivered-payload sequence rather than a fixed-interval time series.

The recursion in eq. 6 is itself backward-looking: $\sigma_t^2$ depends only on $\epsilon_{t-1}^2$ and $\sigma_{t-1}^2$, both determined by information available at or before slot $t-1$. However, the parameters $\hat{\omega}, \hat{\alpha}, \hat{\beta}, \hat{\mu}, \hat{\nu}$ are estimated via maximum likelihood over the full 1,271,078-observation sample, so the fitted recursion at any given slot reflects information from the entire sample period, including slots occurring later in time. This full-sample estimation is appropriate for the descriptive characterization of volatility dynamics undertaken here but the resulting $\hat{\alpha} + \hat{\beta} = 0.980$ should not be interpreted as a quantity that would have been observable or usable in real time without re-estimation.

### 4.3 Market Concentration

Builder identity for each delivered payload was determined from the `builder_pubkey` field returned by each relay's `delivered_payloads` endpoint. Payloads were grouped by raw pubkey rather than by entity-level labels, since not all six relays expose a consistent `extra_data` or builder-name field. This means that distinct pubkeys operated by a single underlying entity are counted as separate competitors; concentration measures reported below should therefore be interpreted as a lower bound on true entity-level concentration.

The Herfindahl-Hirschman Index was computed weekly:

$$\mathrm{HHI}_w = 10{,}000 \times \sum_i s_{i,w}^2 \tag{7}$$

where $s_{i,w}$ is builder $i$'s share of delivered blocks in week $w$. The Gini coefficient was computed both for the full sample (Figure 4) and on a rolling weekly basis (Figure 3), using the trapezoidal Lorenz-curve formula:

$$G = 1 - \sum_{k=1}^{n} (X_k - X_{k-1})(Y_k + Y_{k-1}) \tag{8}$$

where builders are indexed $1, \ldots, n$ in ascending order of blocks produced, $X_k = k/n$ is the cumulative share of builders, and $Y_k$ is the cumulative share of blocks produced by the $k$ lowest-volume builders.

### 4.4 Temporal Pattern Analysis

Systematic temporal patterns were tested using the Kruskal-Wallis test (eq. 4) separately for hour-of-day (24 groups) and day-of-week (7 groups), using the slot-derived UTC timestamps from eq. 1.

### 4.5 Auction Microstructure Analysis

For each matched slot $s$, we compute the following auction statistics from the set of observed bids $\mathcal{B}_s$:

- **Bid count** $|\mathcal{B}_s|$: total number of bids observed across all relays and source types.
- **Builder count** $b_s$: number of distinct builder public keys submitting bids in slot $s$.
- **Relay count** $r_s$: number of distinct relays through which bids were received.

- **Top bid** $v_s^{(1)} = \max_{b \in \mathcal{B}_s} v_b$: highest observed bid value in Wei.
- **Second-highest bid** $v_s^{(2)}$: second-highest observed bid value.
- **Bid margin**:

$$m_s = \frac{v_s^{(1)} - v_s^{(2)}}{v_s^{(1)}} \tag{9}$$

  Under a perfectly competitive second-price auction, $m_s \to 0$; a positive margin indicates that the winning builder faced limited competition.

- **Bid timing**: earliest bid arrival time $\tau_s^{\min}$ and latest bid arrival time $\tau_s^{\max}$, measured in milliseconds relative to slot start ($\tau = 0$). Negative values indicate bids submitted during the prior slot.

Slots with only a single observed bid are classified as *monopoly slots* and excluded from bid margin calculations but retained in participation analyses. We test the association between block value and each auction metric using Spearman's rank correlation, consistent with the non-parametric approach used throughout the paper.

# 5 Empirical Results

## 5.1 Volatility Regime Effects on Block Values

Table 3 presents distributional statistics for block values across the three volatility regimes. Mean block value increases monotonically from 0.0164 ETH in the low-volatility regime to 0.1149 ETH in the high-volatility regime, a 7.0× multiple.

Table 3: Block value (ETH) statistics by volatility regime, September 2024 – May 2026

| Regime | Mean | Median | Std Dev | P25 | P75 | P95 |
|---|---|---|---|---|---|---|
| Low Vol ($n = 419{,}357$) | 0.0164 | 0.0102 | 0.0270 | 0.0070 | 0.0164 | 0.0427 |
| Mid Vol ($n = 432{,}065$) | 0.0313 | 0.0169 | 0.0746 | 0.0107 | 0.0293 | 0.0858 |
| High Vol ($n = 419{,}357$) | 0.1149 | 0.0276 | 2.6463 | 0.0156 | 0.0547 | 0.2136 |

The Kruskal-Wallis test strongly rejects identical distributions across regimes ($H = 266{,}905.3$, $p < 10^{-300}$, $df = 2$). The Mann-Whitney test confirms the directional ordering ($p < 10^{-300}$, one-sided). Spearman's rank correlation is $\rho = 0.494$ ($p < 10^{-300}$, $n = 1{,}270{,}779$), substantially higher than the 5-month subsample estimate ($\rho = 0.322$), confirming the relationship strengthens over longer periods spanning diverse market conditions.

To confirm stability, we bootstrapped the Spearman correlation across 1,000 random samples of $n = 8{,}000$ observations; the mean bootstrap $\hat{\rho} = 0.493$ with std. dev. = 0.009 and 95% CI [0.477, 0.510], confirming the estimate is stable and not sensitive to any particular subsample.

> **Finding 1.** *Block values are approximately 7.0× higher on average during high-volatility regimes relative to low-volatility regimes ($H = 266{,}905$, $\rho = 0.494$, $p < 10^{-300}$). The effect is substantially stronger than estimates from shorter sample periods, reflecting the inclusion of significant market stress events across the 20-month sample.*

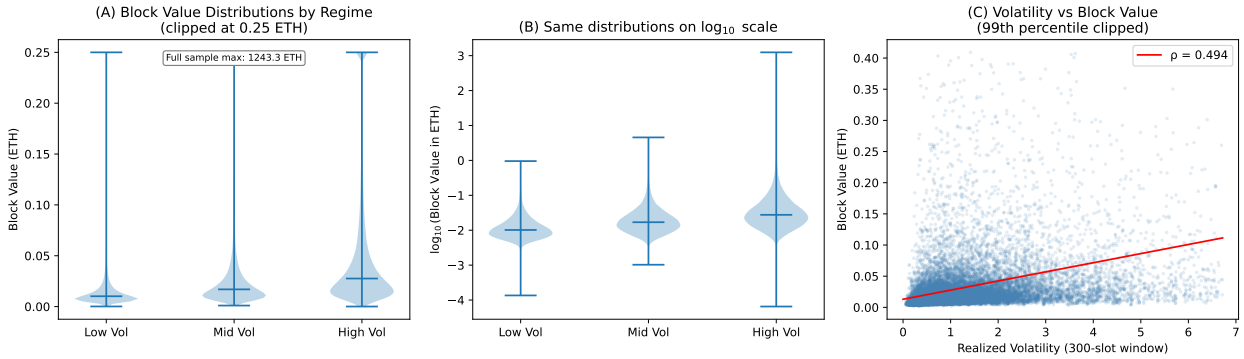



Figure 1: (A) Block value distributions across volatility regimes, clipped at 0.25 ETH for readability. (B) Same distributions on a $\log_{10}$ scale showing the full range. (C) Scatter of realized volatility against block value with fitted linear trend. Full-sample Spearman $\rho = 0.494$ ($p < 10^{-300}$); bootstrap mean $\hat{\rho} = 0.493$ (95% CI: [0.477, 0.510], $B = 1{,}000$ samples).

## 5.2 GARCH Dynamics of MEV Block Values

Table 4 reports Ljung-Box test results on squared log returns.

Table 4: Ljung-Box test for ARCH effects in squared log returns ($n = 1{,}271{,}078$)

| Lags | Q-statistic | $p$-value |
|---|---|---|
| 10 | 257,972.5 | $< 10^{-300}$ |
| 20 | 268,977.5 | $< 10^{-300}$ |
| 50 | 280,427.9 | $< 10^{-300}$ |

Table 5 presents the GARCH(1,1) parameter estimates. The estimated volatility persistence is $\hat{\alpha} + \hat{\beta} = 0.100 + 0.880 = 0.980$, indicating extremely high but non-unit-root volatility persistence.

Table 5: GARCH(1,1) parameter estimates with Student's $t$ errors, full 20-month sample

| Parameter | Estimate | Std Error | $t$-stat | $p$-value |
|---|---|---|---|---|
| $\mu$ (mean) | $-5.59 \times 10^{-6}$ | 0.00117 | $-0.00$ | 0.996 |
| $\omega$ (constant) | 2.5788 | 0.00476 | 541.4 | $< 0.001$ |
| $\alpha$ (ARCH) | 0.1000 | 0.000662 | 151.0 | $< 0.001$ |
| $\beta$ (GARCH) | 0.8800 | 0.000183 | 4797.7 | $< 0.001$ |
| $\nu$ (d.o.f.) | 4.0053 | 0.00152 | 2643.6 | $< 0.001$ |

$\hat{\alpha} + \hat{\beta} = 0.980$; AIC = 7,067,110; $n = 1{,}271{,}078$; converged

**Finding 2.** *MEV block value log returns exhibit highly persistent volatility dynamics ($\hat{\alpha} + \hat{\beta} = 0.980$) with very heavy tails ($\hat{\nu} = 4.005$), confirmed across GARCH(1,1) and winsorised specifications. A persistence of 0.980 implies that volatility shocks retain approximately 82% of their impact after ten periods. Subperiod GARCH estimation is not reliable for this series due to the non-standard distribution of* $\log(1 + v_t)$ *returns at sub-annual horizons.*

Table 6: GARCH robustness checks: volatility persistence ($\hat{\alpha} + \hat{\beta}$)

| Specification | Sample | Persistence | Note |
|---|---|---|---|
| GARCH(1,1) | Full 20-month | 0.9800 | Stable |
| GARCH(1,1), winsorized 1% | Full 20-month | 0.9800 | Stable |
| GARCH(1,1) | Subperiods | – | Non-convergent |

Subperiod GARCH does not converge reliably due to the mixture-like distribution of $\log(1 + v_t)$ returns at sub-annual horizons. The full-sample estimate is the most reliable characterisation.

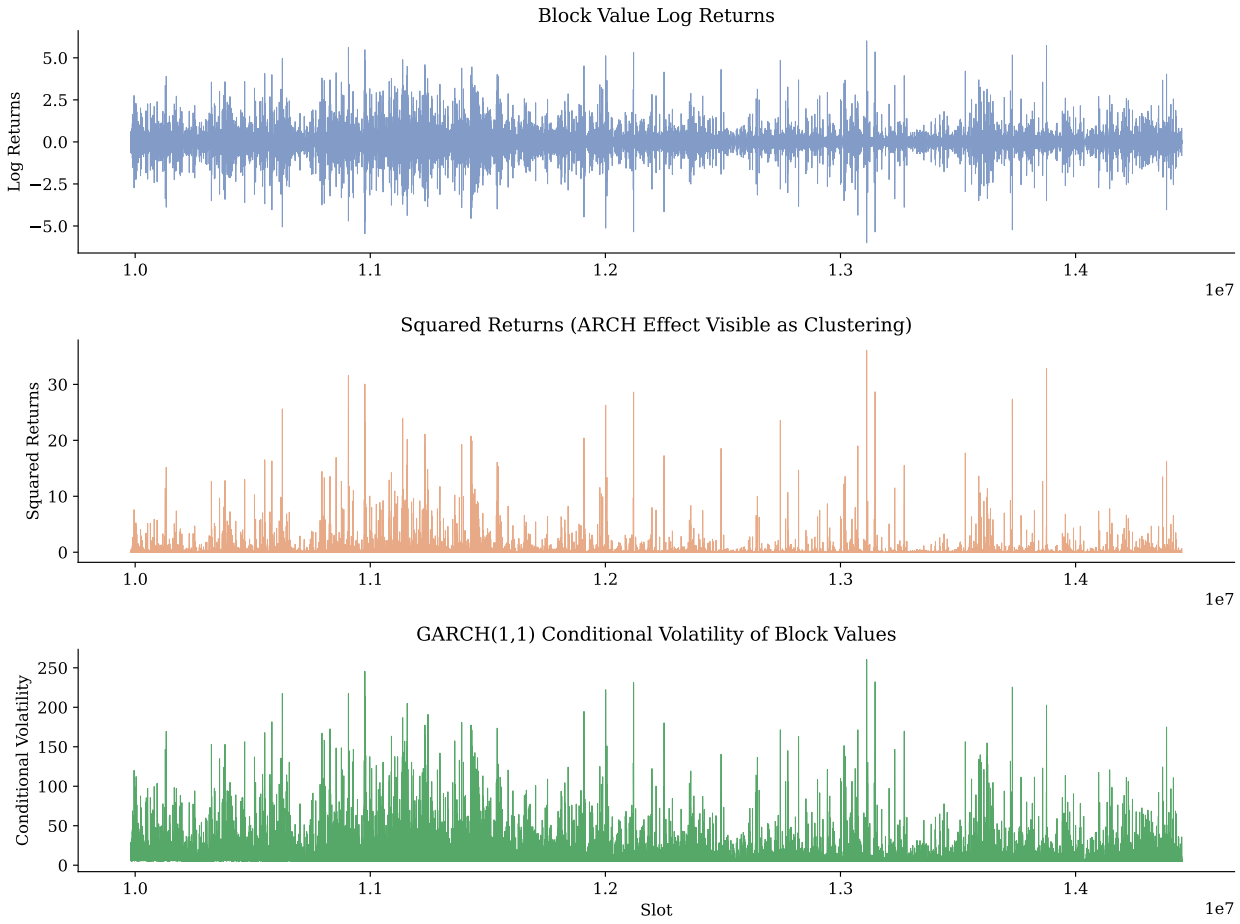



Figure 2: Top: Log returns of block values across 1,271,078 slots. Middle: Squared returns, illustrating ARCH-type volatility clustering. Bottom: GARCH(1,1) conditional volatility estimate $\hat{\sigma}_t$.

### 5.3 Builder Market Concentration

Figure 3 displays the weekly HHI and Gini coefficient over the full 20-month sample. The HHI began at approximately 2,100 in September 2024 and declined sharply to approximately 800–900 by early 2025. A secondary period of elevated concentration occurred in mid-2025 (HHI briefly reaching approximately 1,500), followed by a return to the unconcentrated range of 600–1,000.

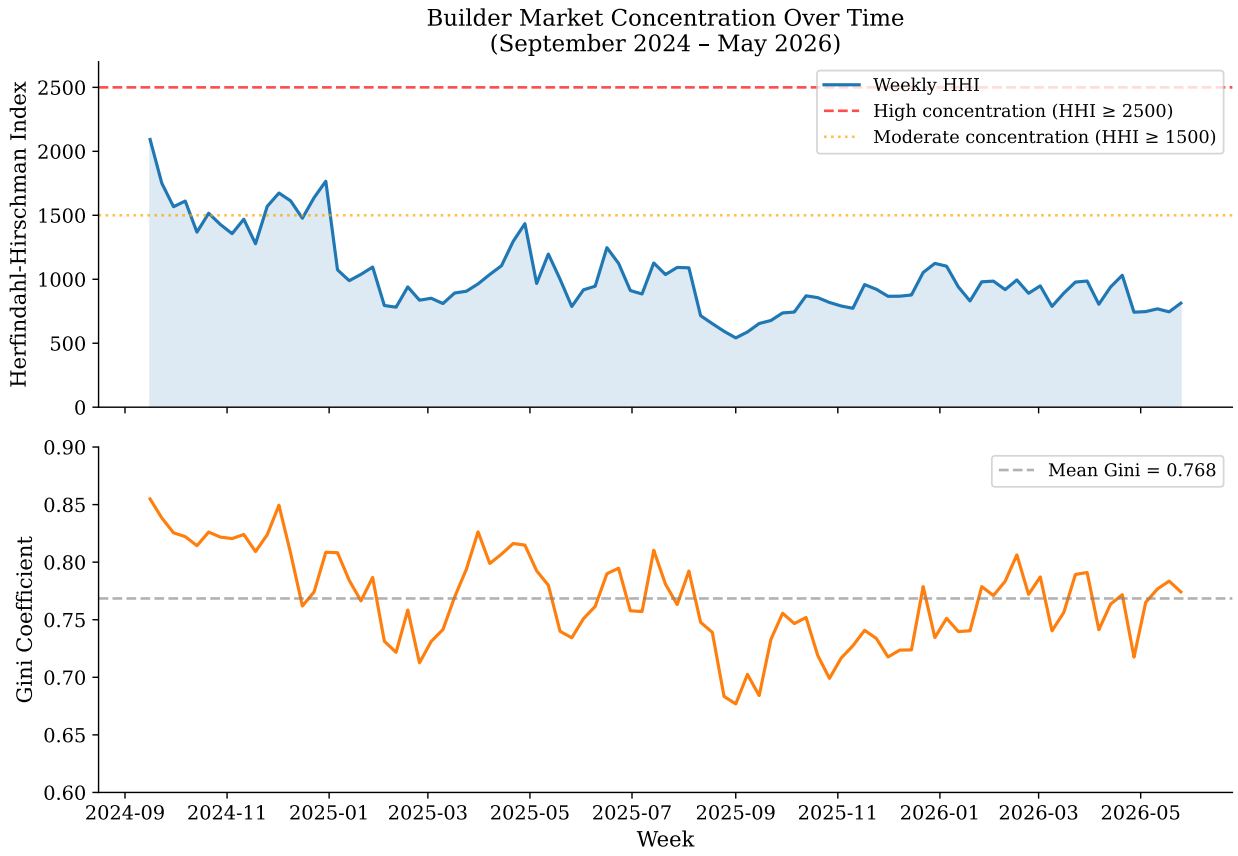



Figure 3: Top: Weekly HHI of builder block production (eq. 7). Bottom: Weekly Gini coefficient (eq. 8). The HHI shows significant initial deconcentration followed by cyclical variation, while the Gini declines gradually from approximately 0.85 to 0.77.

**Finding 3.** *Builder HHI declined from approximately 2,100 in September 2024 to a stable unconcentrated range of 600–1,000, with a secondary concentration episode in mid-2025 (HHI* $\approx$ 1,500*). The Gini coefficient declined from 0.85 to a long-run mean of 0.77.*

### 5.4 Temporal Patterns in MEV Activity

Figure 5 displays mean block values by hour of day (UTC) and day of week. The Kruskal-Wallis test rejects equal distributions across hours ($H = 65{,}875$, $df = 23$, $p < 10^{-300}$) and across days of the week ($H = 33{,}590$, $df = 6$, $p < 10^{-300}$). Block values are highest on Monday (mean 0.086 ETH) and lowest on Saturday (0.031 ETH). The intraday pattern is clearest in medians: median block value rises from approximately 0.012–0.013 ETH during 03:00–05:00 UTC to 0.020–0.024

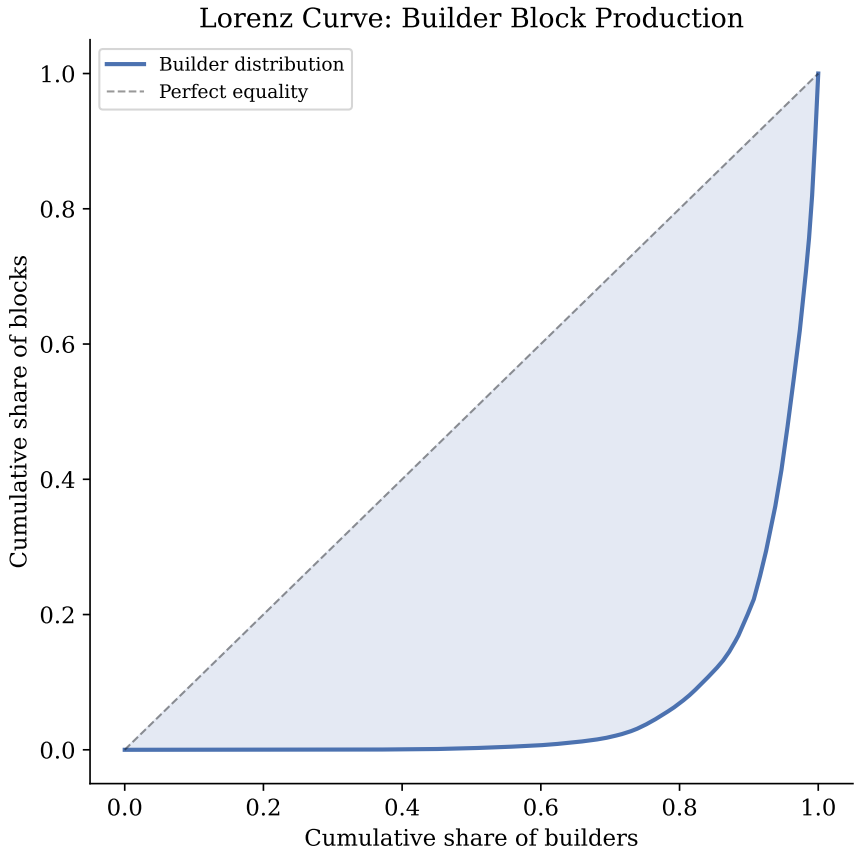



Figure 4: Lorenz curve of builder block production shares, full 20-month sample. Pooled Gini = 0.868 across 227 builders: the bottom 50% of builders account for approximately 0.2% of delivered blocks. The pooled full-sample Gini exceeds the weekly-average Gini of 0.77 reported in Figure 3 because pooling over 20 months captures builders that were active only briefly.

ETH during 13:00–17:00 UTC, consistent with elevated activity during US trading hours. Hourly means are dominated by a small number of extreme blocks (the 02:00 UTC mean, for example, reflects a single outlier with hourly standard deviation 5.5 ETH) and should be read with that caveat.

> **Finding 4.** *MEV block values exhibit statistically significant intraday and day-of-week seasonality ($p < 10^{-300}$ in both cases). Median block values peak during 13:00–17:00 UTC and are highest on Monday and lowest on Saturday.*

## 5.5 Auction Microstructure

### 5.5.1 Overall Competitiveness

Table 7 presents summary statistics for the auction dataset across 1,220,100 matched slots. The PBS auction is highly competitive at the median: builders submit a median of 132 bids per slot from 11 distinct participants across 3 relays. Fewer than 0.001% of slots exhibit a monopoly configuration, indicating near-universal contestability.

The distribution of bid margins is highly right-skewed: the median of 0.141% implies near-perfect competition in typical slots, while the 90th-percentile margin of 6.691% indicates that one

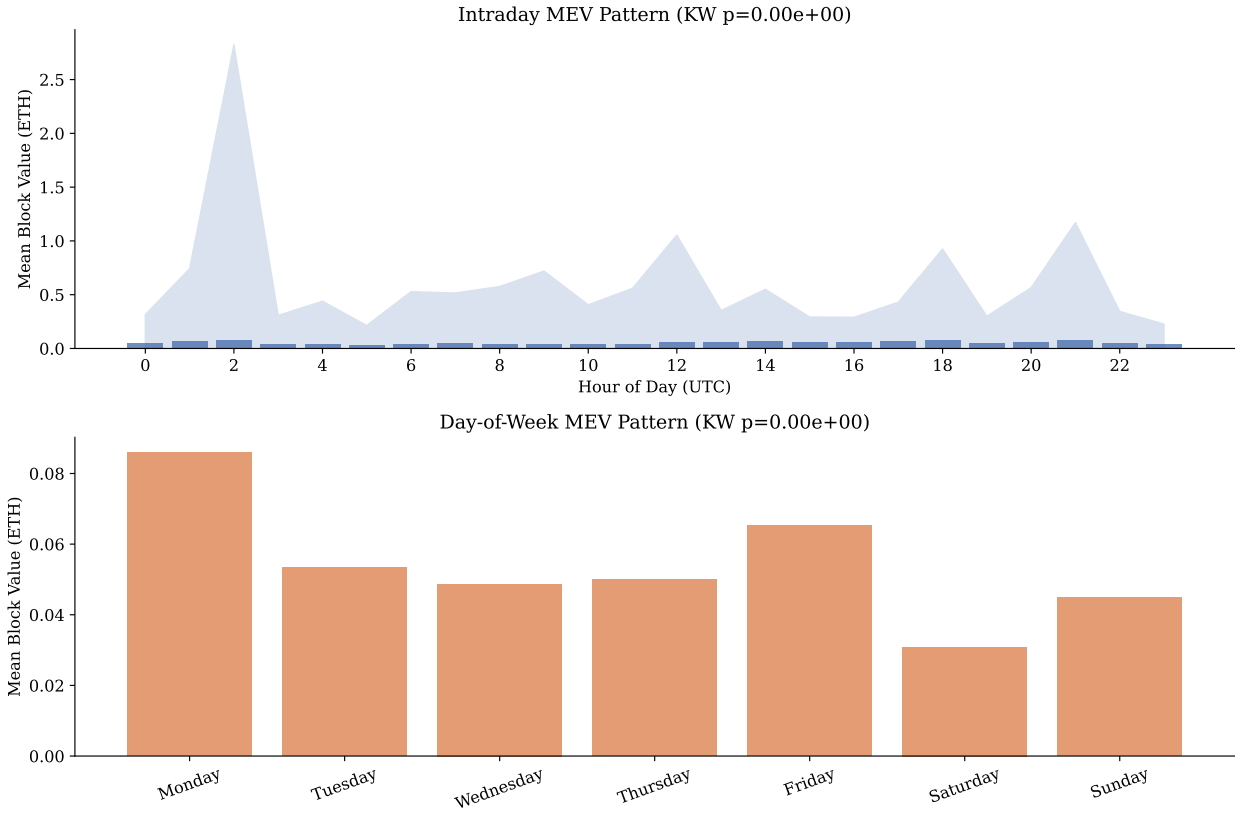



Figure 5: Top: Mean block value by hour of day (UTC); isolated spikes reflect individual extreme blocks rather than systematic patterns. Bottom: Mean block value by day of week. Both panels show statistically significant variation ($p < 10^{-300}$).

in ten slots exhibits materially reduced competition. Figure 6 displays the full distributions of bid counts, builder participation, and bid margins.

### 5.5.2 Competitiveness Increases with Block Value

Figure 7 displays the relationship between block value and bid margin. Bid margins decline monotonically as block value increases, with median margins of 0.196%, 0.144%, and 0.110% for low-, mid-, and high-value terciles respectively. Spearman's rank correlation between block value and bid margin is $\rho = -0.054$ ($p < 10^{-300}$, $n$ = 1,219,851; bootstrap 95% CI $[-0.075, -0.031]$, $B$ = 1,000). Builder participation exhibits a stronger positive association with block value ($\rho = 0.286$, $p < 10^{-300}$; bootstrap 95% CI $[0.265, 0.305]$): high-value slots attract substantially more builders.

The compression of bid margins in high-value slots provides an auction-theoretic mechanism underlying the volatility-value correlation documented in Finding 1: in the slots where MEV opportunity is largest, proposers capture a larger share of that value because competition drives bids toward the true block value.

> **Finding 5a.** *Bid margins compress monotonically with block value (0.196%, 0.144%, 0.110% across low, mid, and high value terciles; $\rho = -0.054$, $p < 10^{-300}$), and builder participation increases with block value ($\rho = 0.286$, $p < 10^{-300}$). High-value slots are both more competitive and more efficiently allocated.*

Table 7: Auction microstructure summary statistics, 1,220,100 matched slots, September 2024 – May 2026

| Metric | Mean | Median | P10 | P90 | Max |
|---|---|---|---|---|---|
| Bids per slot | 141.5 | 132 | 19 | 251 | 7,135 |
| Builders per slot | 11.5 | 11 | 7 | 17 | 37 |
| Relays per slot | 3.0 | 3 | 2 | 5 | 8 |
| Bid margin (%) | 3.265 | 0.141 | 0.000 | 6.691 | 100.00 |
| Earliest bid (ms) | −7,916 | −8,090 | −11,018 | −3,898 | 0 |
| Latest bid (ms) | 2,187 | 2,058 | 629 | 3,647 | 22,082 |

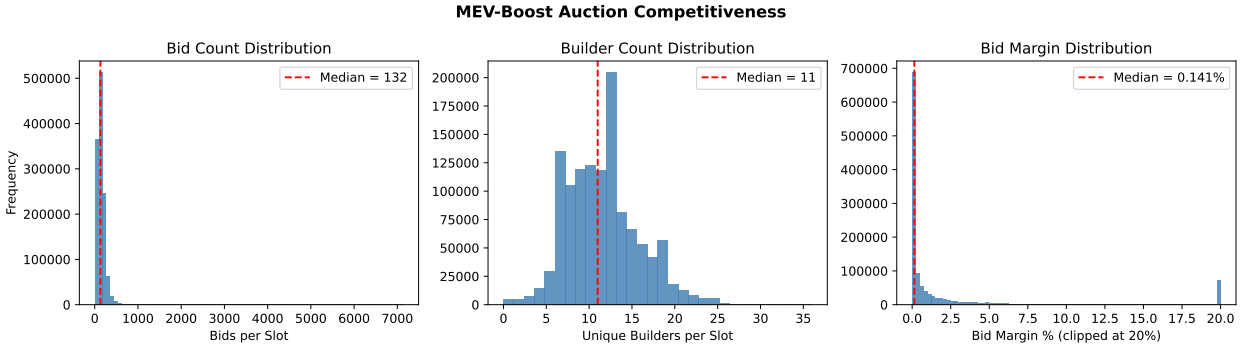



Figure 6: Distribution of auction competitiveness metrics across 1,220,100 matched slots. Left: bid count per slot (median 132). Centre: unique builders per slot (median 11). Right: bid margin distribution, clipped at 20% for readability (median 0.141%).

### 5.5.3 Bid Timing

Figure 8 displays the distribution of earliest and latest bid arrival times. Builders begin submitting bids a median of 8,090 ms before slot start, indicating that bidding activity commences during the preceding slot's window. Latest bid arrivals are concentrated around 2,058 ms after slot start, with 5.3% of slots receiving bids more than 4,000 ms after slot start.

Figure 9 shows that high-value slots attract later bids ($\rho = 0.146$, $p < 10^{-300}$; bootstrap 95% CI $[0.125, 0.169]$) and more builders. Builders extend their bidding window when the potential reward is larger, consistent with rational cost-benefit behaviour.

> **Finding 5b.** *Builders begin bidding a median of 8,090 ms before slot start and continue until a median of 2,058 ms after slot start. High-value slots attract later bids ($\rho = 0.146$, $p < 10^{-300}$) and more builders ($\rho = 0.286$, $p < 10^{-300}$), consistent with rational extension of the bidding window in pursuit of higher expected payoffs.*

### 5.5.4 Temporal Evolution of Auction Competitiveness

Figure 10 displays weekly medians of bid margin, builder participation, and bid count over the full 20-month sample. None of the three series exhibits the monotone "deepening" that a naive maturation narrative would predict. Median builder participation *declined* on net, from approximately 17 builders per slot in late 2024 to approximately 9–11 by 2026. Median bid counts are cyclical without a clear trend, fluctuating between roughly 75 and 225 per slot around a full-sample median

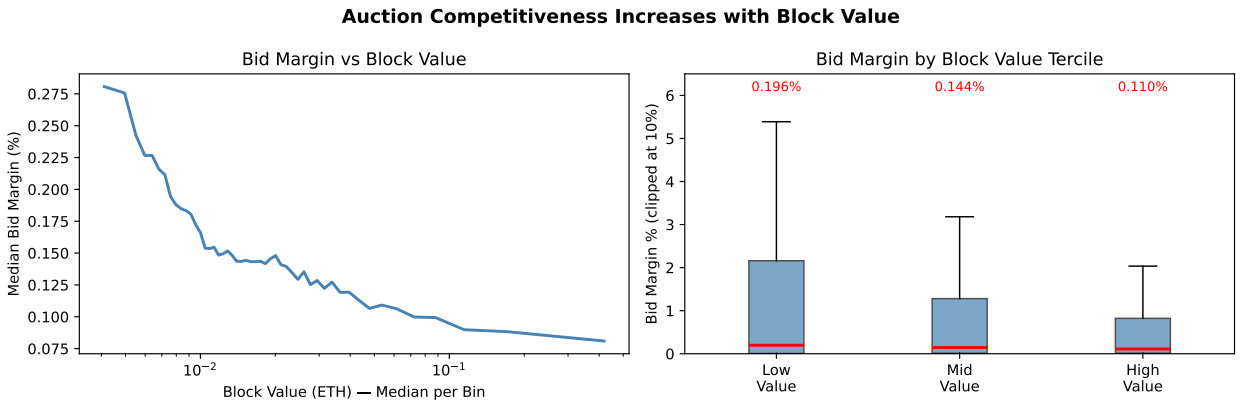



Figure 7: Left: scatter of bid margin against block value (log scale) with rolling median (red line). Right: bid margin distributions by block value tercile (outliers removed; median annotated in red). Bid margins compress monotonically with block value, consistent with intensified competition for high-MEV slots.

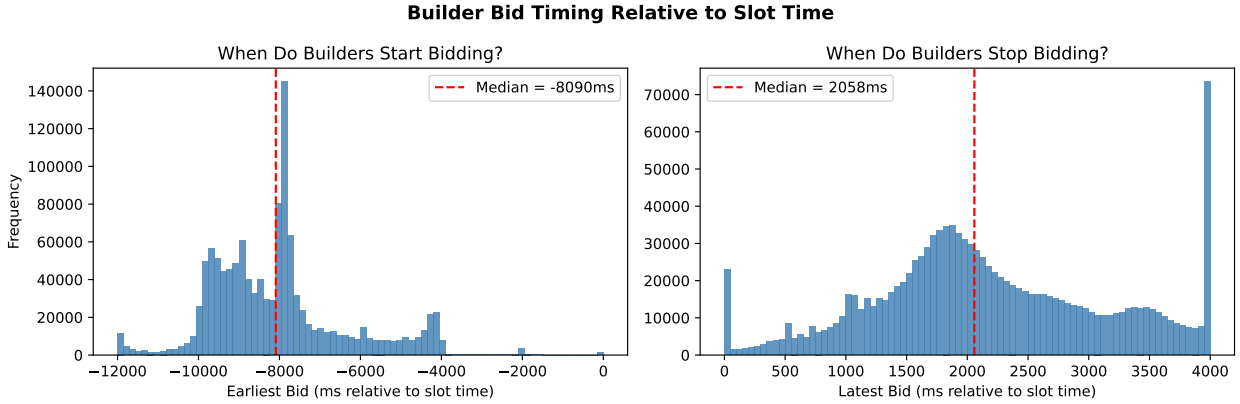



Figure 8: Builder bid timing relative to slot start ($\tau = 0$), across 1,220,100 matched slots. Left: earliest bid arrival (median −8,090 ms). Right: latest bid arrival (median +2,058 ms); 5.3% of slots receive bids after +4,000 ms.

of 132. Median bid margins show no sustained compression: they oscillate between approximately 0.05% and 0.3% through 2025 and are, if anything, elevated (0.2–0.5%) during December 2025 – March 2026. Monthly matched-slot counts also jump roughly five-fold during January–April 2025 relative to adjacent months, reflecting uneven bid-archive coverage rather than a change in auction activity; the weekly medians are robust to this because they condition on matched slots, but level comparisons across that boundary should be made with care.

Read against the HHI deconcentration documented in Section 5.3, this is a picture of consolidation rather than entry: the number of builders actively bidding per slot fell by roughly 40% while the distribution of *delivered* blocks among the remaining builders became more even. Fewer, more evenly matched competitors can produce both a lower HHI and stable margins, which is what the data show.

Table 8 formalises these bivariate relationships in a multivariate setting, regressing bid margin jointly on log block value, builder count, relay count, hour of day, and day of week. All five predictors remain statistically significant ($p < 0.001$) with the expected signs after controlling for

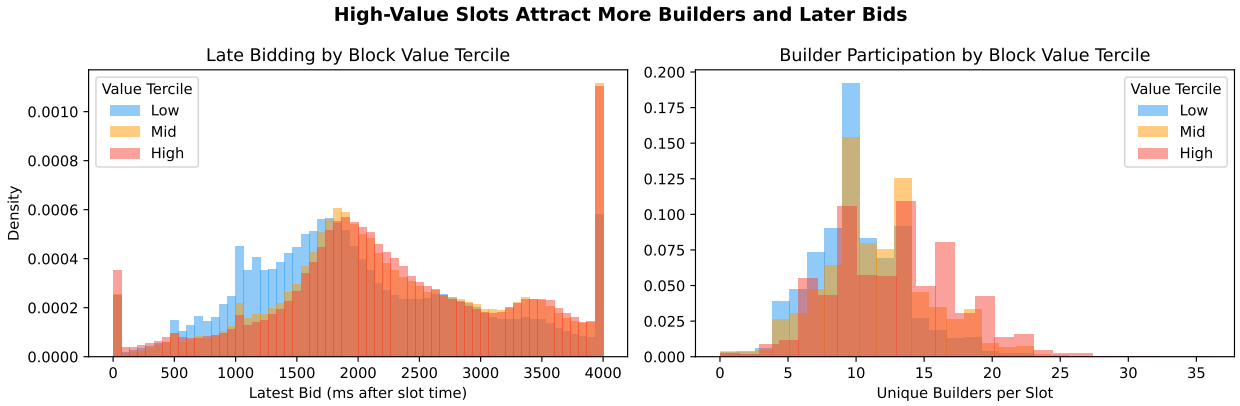



Figure 9: Left: distribution of latest bid arrival time by block value tercile. High-value slots exhibit a rightward shift, indicating later bidding when rewards are larger. Right: builder participation distributions by value tercile.

the others: a one-unit increase in log block value is associated with a 0.76 percentage point decline in margin, each additional builder with a 0.16 percentage point decline, and each additional relay with a 0.72 percentage point decline — relay participation has the largest independent effect on margin compression among the three auction-structure variables. The model explains a modest but significant share of margin variance ($R^2 = 0.020$), consistent with bid margins being driven by many factors beyond the five auction-structure variables considered here, including builder-specific cost structures and idiosyncratic orderflow access documented in Section 5.6.4.

> **Finding 5c.** *The PBS auction has consolidated rather than deepened: median builder participation declined from approximately 17 to 9–11 builders per slot over the sample, while median bid counts (75–225) and bid margins (0.05–0.5%) fluctuate cyclically without a sustained trend. Combined with the HHI deconcentration in delivered blocks, this indicates a smaller but more evenly matched set of active builders.*

## 5.6 Extended Auction Analysis

### 5.6.1 ETH Price Decomposition

A key limitation of prior PBS analyses is the conflation of ETH price movements with MEV-specific dynamics. We address this directly using 15,302 hourly ETH/USD price observations from Coinbase (September 2024–May 2026), matched to delivered payload slots by hour.

Of the 1,271,079 delivered payload slots, 1,218,882 (95.9%) are matched to an hourly ETH price observation; the unmatched remainder corresponds to hours for which no Coinbase price bar was retrieved. We regress log block value (ETH) on log ETH price and find:

$$\log(v_t) = -4.172 + 0.027 \cdot \log(P_t^{\text{ETH}}) + \varepsilon_t, \quad R^2 = 0.0001 \tag{10}$$

where $P_t^{\text{ETH}}$ denotes the hourly ETH/USD closing price. ETH price explains essentially zero variance in block value when both are denominated in ETH (Spearman $\rho = 0.050$, $p < 10^{-300}$). However, when block value is converted to USD, the association strengthens substantially (Spearman $\rho = 0.359$, $p < 10^{-300}$), reflecting the mechanical co-movement of any ETH-denominated quantity with its USD equivalent.

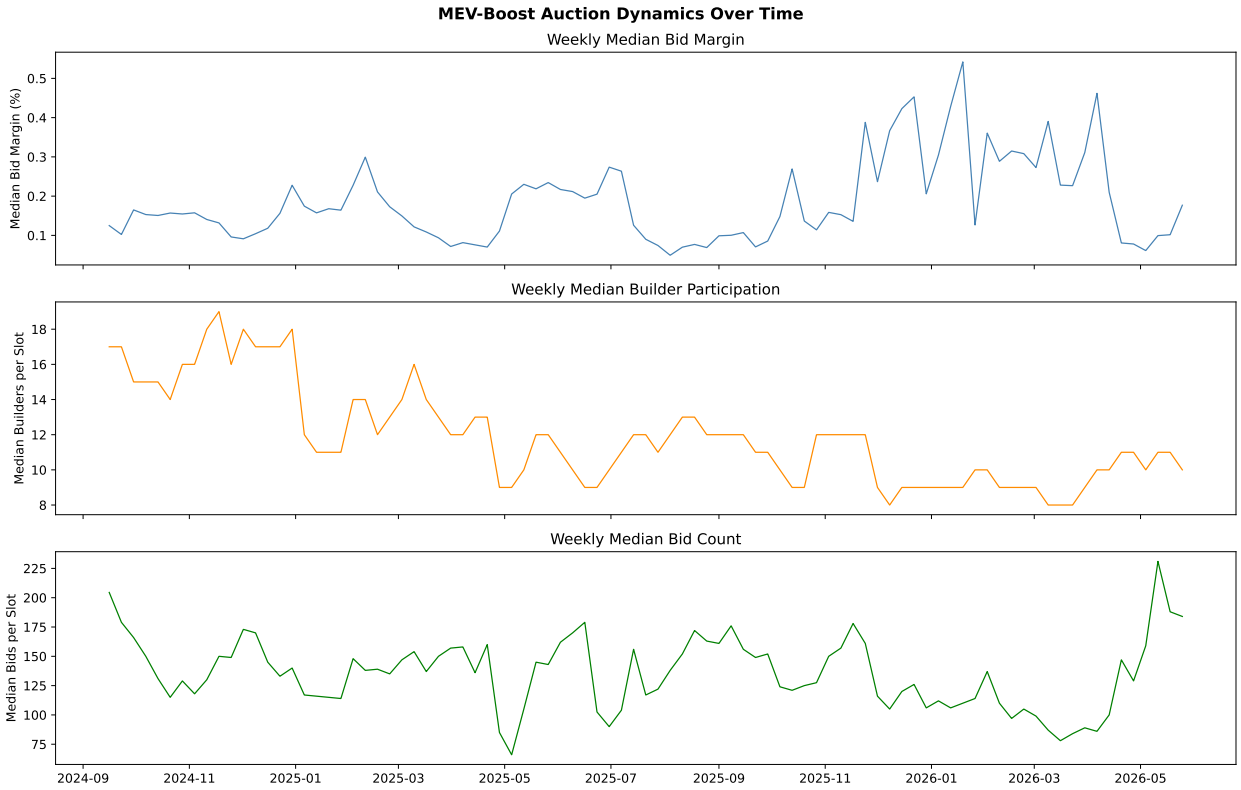



Figure 10: Weekly medians of auction competitiveness metrics, September 2024 – May 2026. Top: median bid margin (no sustained trend; elevated in late 2025 – early 2026). Middle: median builder participation (declining from approximately 17 to 9–11 per slot). Bottom: median bid count (cyclical, no clear trend).

These results establish that MEV block value in ETH is driven by on-chain activity—transaction ordering opportunities, arbitrage spreads, and liquidation events—rather than by ETH price movements. The volatility regime effects documented in Finding 1 (7.0× block value increase in high-volatility periods) therefore reflect genuine variation in on-chain MEV opportunity, not a repricing artifact of ETH/USD fluctuations.

> **Finding 6.** *ETH price explains essentially zero variance in block value when denominated in ETH ($R^2 = 0.0001$, Spearman $\rho = 0.050$), but explains meaningful variance in USD terms ($\rho = 0.359$). MEV block value is driven by on-chain activity, not ETH price movements. The 7.0× volatility regime effect documented in Finding 1 reflects genuine variation in MEV opportunity.*

### 5.6.2 Isolating MEV-Specific Volatility Persistence

To formally confirm that the GARCH persistence documented in Finding 2 is driven by MEV-specific dynamics rather than ETH price movements, we regress the block-value log return series on the contemporaneous ETH price log return and re-estimate GARCH(1,1) on the residual. The regression yields $R^2 \approx 0.0000$ ($n = 1{,}269{,}860$), confirming that ETH price returns explain essentially none of the variance in block-value returns at the slot level, consistent with Finding 6.

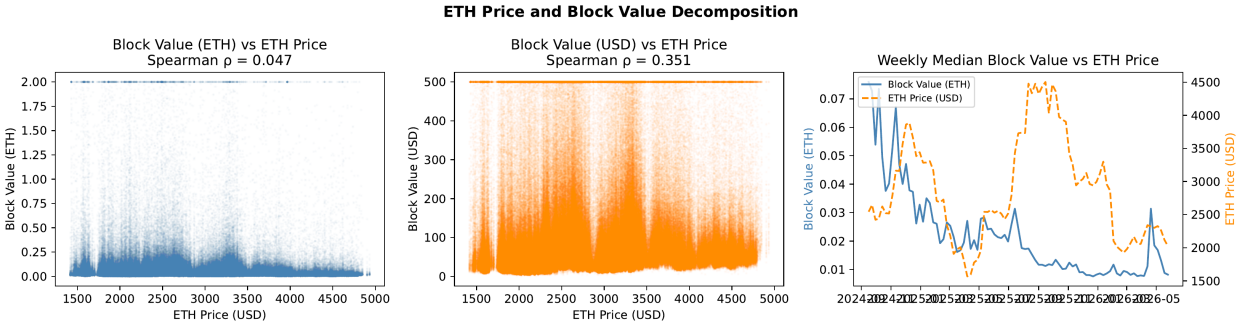



Figure 11: ETH price and block value decomposition. Left: block value in ETH vs ETH price (Spearman $\rho = 0.050$); MEV opportunity is orthogonal to ETH price in ETH terms. Centre: block value in USD vs ETH price ($\rho = 0.359$), reflecting mechanical USD conversion. Right: weekly median block value (ETH, blue) and ETH price (USD, orange) co-plotted, confirming the visual decoupling in ETH terms.

Table 8: OLS regression of bid margin on auction structure variables ($n$ = 1,219,851, HC0 robust standard errors)

| Variable | Coefficient | Robust SE | $t$-stat | $p$-value |
|---|---|---|---|---|
| Intercept | 3.7968 | 0.0620 | 61.24 | <0.001 |
| log(Block Value) | −0.7576*** | 0.0092 | −82.62 | <0.001 |
| N Builders | −0.1564*** | 0.0024 | −66.24 | <0.001 |
| N Relays | −0.7190*** | 0.0078 | −92.56 | <0.001 |
| Hour of Day | 0.0080*** | 0.0013 | 6.01 | <0.001 |
| Day of Week | 0.0119* | 0.0047 | 2.51 | 0.012 |

$R^2 = 0.020$. $^{***}p < 0.001$, $^{*}p < 0.05$.

Dependent variable: bid margin (%). Each additional builder reduces the margin by 0.16 percentage points; each additional relay reduces it by 0.72 percentage points, holding other variables constant.

GARCH(1,1) estimated on this MEV-specific residual series yields $\hat{\alpha} + \hat{\beta} = 0.980$ ($\hat{\nu} = 4.005$), numerically identical to the full block-value return series. This confirms that the high volatility persistence documented in Finding 2 is a property of MEV dynamics in isolation, not an artifact of correlated ETH price movements. The PBS auction's volatility clustering is therefore a genuinely on-chain phenomenon, reinforcing the interpretation in Section 5.6.1 that MEV extraction opportunity is fundamentally driven by transaction-level activity rather than asset price dynamics.

### 5.6.3 Relay Participation as a Determinant of Auction Depth

We examine whether the number of relays through which bids are received per slot predicts auction competitiveness. Table 9 presents median bid margins by relay count.

Bid margins decline monotonically from 0.217% in single-relay slots to 0.066% in 7-relay slots, a 3.3× compression. Spearman's rank correlation confirms the relationship (excluding the two sparse extreme categories): $\rho = -0.057$, $p < 10^{-300}$. While the effect size is modest, the monotone pattern across seven relay-count categories and the statistical significance across 1.2M observations establish relay participation as a structural determinant of auction depth. Slots reaching more relays expose builders to wider competition, driving margins lower.

**Finding 7.** *Relay participation is a structural determinant of auction competitiveness:*

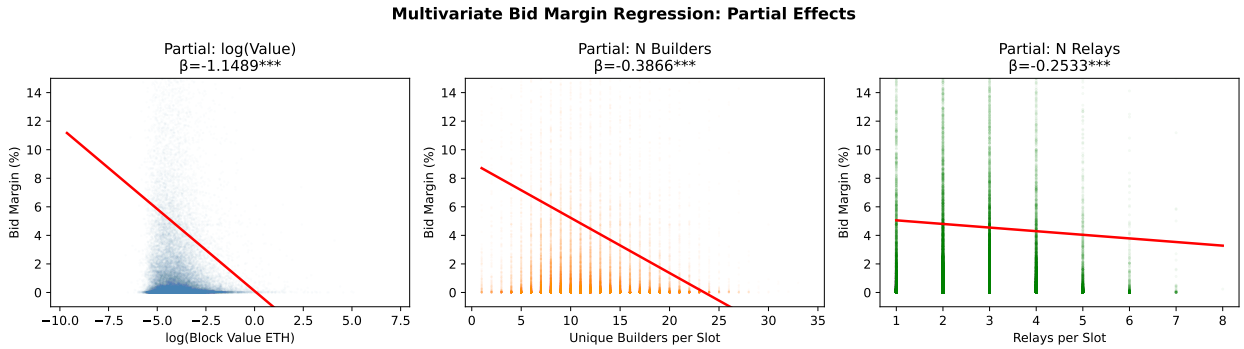



Figure 12: Partial regression plots for the three main auction structure variables. Red lines show OLS fitted values at mean levels of all other regressors. All three variables are statistically significant ($p < 0.001$) with the expected negative sign, confirming that block value, builder participation, and relay participation independently reduce bid margins.

Table 9: Bid margin by relay count per slot

| Relays per Slot | Median Bid Margin (%) | Slot Count |
|---|---|---|
| 1 | 0.217 | 213,316 |
| 2 | 0.143 | 401,220 |
| 3 | 0.124 | 350,993 |
| 4 | 0.124 | 195,353 |
| 5 | 0.109 | 49,090 |
| 6 | 0.084 | 8,714 |
| 7 | 0.066 | 1,112 |

*bid margins decline monotonically from 0.217% in single-relay slots to 0.066% in 7-relay slots ($\rho = -0.057$, $p < 10^{-300}$; bootstrap 95% CI $[-0.079, -0.035]$, $B = 1{,}000$). Broader relay participation exposes builders to wider competition, compressing margins by up to 3.3×.*

### 5.6.4 Builder Strategy Heterogeneity

Analysis of 5,130,040 bid observations across 144 builders in a Q1 2025 weekly sample initially suggested three bidding clusters, including a small group of builders whose bids arrived after slot start with markedly higher median values. To test whether this reflected genuine builder strategy or an artifact of relay-side data collection, we reprocessed the complete (non-sampled) Q1 2025 bid archive — 90 daily files, 86,805,657 bid observations — and compared bid timing for the same builders across different relays.

The results show that bid timing is overwhelmingly a property of the *relay*, not the builder. Pooling all builders, median bid timing ranges from −4,913 ms (securerpc) to +2,497 ms (Titan) depending on relay, a spread of nearly 7.4 seconds driven entirely by relay-specific data collection and reporting characteristics; ultrasound alone contributes 96.3% of all observations with a median of −710 ms, while bloxroost's two endpoints report medians of +2,104 to +2,208 ms. Critically, when the same builder's bids are compared across relays, timing shifts by 500–3,000 ms depending on which relay reported them (e.g. builder `0xb4a435...` shows a median of +2,135 ms via bloxroute

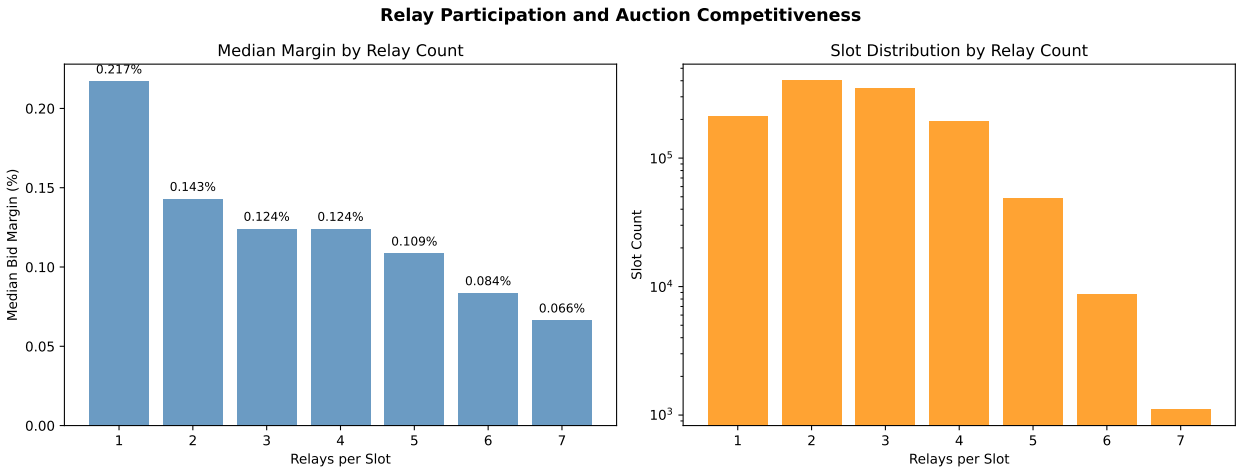



Figure 13: Left: median bid margin by relay count per slot. Margins decline monotonically from 0.217% (1 relay) to 0.066% (7 relays). Right: slot distribution by relay count; 2–3 relays account for the majority of slots.

versus +1,586 ms via other relays). The apparent "late bidding" cluster in the weekly sample was therefore an artifact of relay composition in that sample rather than a distinct builder strategy, and is not reported as a separate cluster. A corrected win-rate analysis using the full 90-day bid archive as the denominator shows these builders have unremarkable win rates (median 0.59%, pooled 8.62%) and winning block values statistically indistinguishable from the market average, ruling out the original hypothesis of exclusive high-value orderflow access.

Two genuine bidding strategies remain after this correction. The first group submits approximately 18 bids per slot with median timing of −6,759 ms, bidding early and conservatively well before slot start, and is associated with lower median bid values (0.009 ETH). The second group submits 2–4 bids per slot with median timing near slot start (−416 ms), bidding closer to the deadline, and is associated with higher median bid values (0.021 ETH).

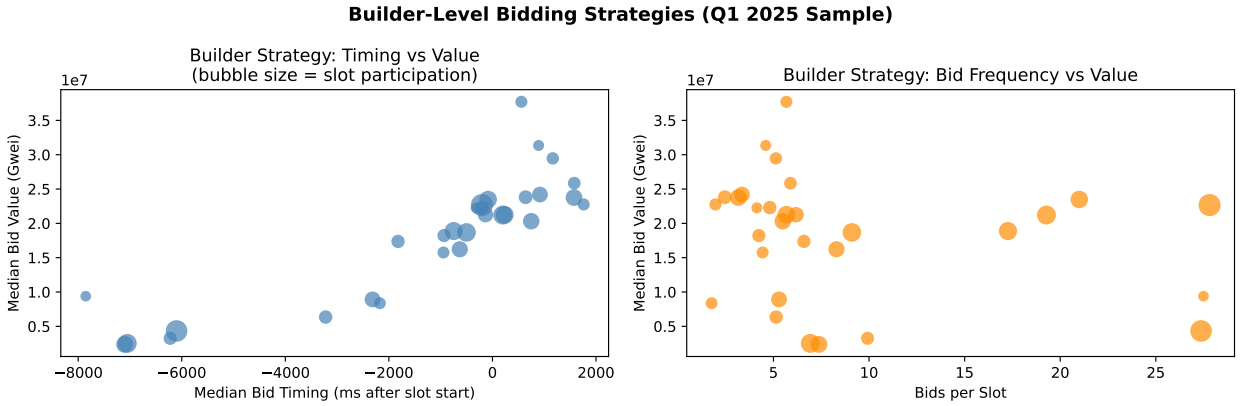



Figure 14: Builder-level bidding strategies (Q1 2025 sample, 144 builders, 5.1M bid observations). Left: median bid timing vs median bid value; bubble size indicates slot participation. Right: bids per slot vs median bid value. Two genuine clusters are apparent: early-conservative high-frequency bidders and mid-frequency near-slot-time bidders.

**Finding 8.** *After correcting for a relay-side timing artifact (Section 5.6.5), builder bidding strategies cluster into two genuine types: early-conservative builders submitting ∼18 bids per slot at −6,759 ms with low median values (0.009 ETH), and mid-frequency builders submitting ∼2–4 bids per slot near slot start (−416 ms) with higher median values (0.021 ETH). An initially apparent third "post-slot-time" cluster was found to be an artifact of relay-specific bid-timestamp reporting rather than a distinct builder strategy (Finding 8b).*

### 5.6.5 Relay-Level Bid Timestamp Heterogeneity

The discovery that an apparent builder strategy cluster was actually a relay reporting artifact motivates a direct examination of timestamp heterogeneity across relays. Table 10 reports the distribution of bid arrival times by relay, pooled across all 86.8 million Q1 2025 bid observations.

Table 10: Bid timing (ms relative to slot start) by relay, full Q1 2025 bid archive ($n$ = 86,805,657)

| Relay | Median (ms) | Mean (ms) | Observations |
|---|---|---|---|
| Titan | 2,497 | 580 | 1,512,420 |
| BloXroute (regulated) | 2,208 | 2,421 | 359,445 |
| BloXroute (max-profit) | 2,104 | 2,303 | 237,888 |
| Agnostic | 1,890 | 1,748 | 125,789 |
| Flashbots | −454 | −346 | 790,105 |
| Ultrasound | −710 | −1,946 | 83,630,070 |
| Aestus | −4,258 | −3,334 | 126,004 |
| Eden Network | −4,447 | −3,548 | 15,169 |

The spread between the latest-reporting relay (Titan, median +2,497 ms) and the earliest-reporting relay (Eden Network, median −4,447 ms) exceeds 6,900 ms — more than half a slot duration. Because Ultrasound's websocket stream contributes 96.3% of all observations and reports systematically earlier timestamps than other relays, any subset of bids drawn disproportionately from non-Ultrasound relays will appear to arrive later in absolute terms, independent of genuine builder behaviour. This pattern most plausibly reflects differences in each relay's bid collection and timestamping pipeline (e.g. polling versus streaming architectures, network propagation delay to the Relayscan collector) rather than differences in when builders actually submit bids to each relay.

**Finding 8b.** *Bid arrival timestamps in the Relayscan archive vary systematically by relay, with a median spread of 6,944 ms between the latest- and earliest-reporting relays (Titan vs. Eden Network), exceeding half a slot duration. This relay-specific heterogeneity, rather than builder strategy, explains an initially apparent "post-slot-time" builder cluster (Finding 8), and implies that cross-relay bid-timing comparisons in MEV-Boost research must condition on relay identity to avoid spurious behavioural inferences.*

### 5.6.6 GARCH Dynamics of Bid Margins

We apply GARCH(1,1) analysis to the bid margin series to characterise volatility dynamics in auction competitiveness. The Ljung-Box test on squared demeaned margins rejects absence of ARCH effects at all lag lengths ($Q_{10}$ = 11,070, $Q_{20}$ = 19,158, $Q_{50}$ = 40,186; all $p < 0.001$), confirming that bid margin volatility is time-varying and clustered.

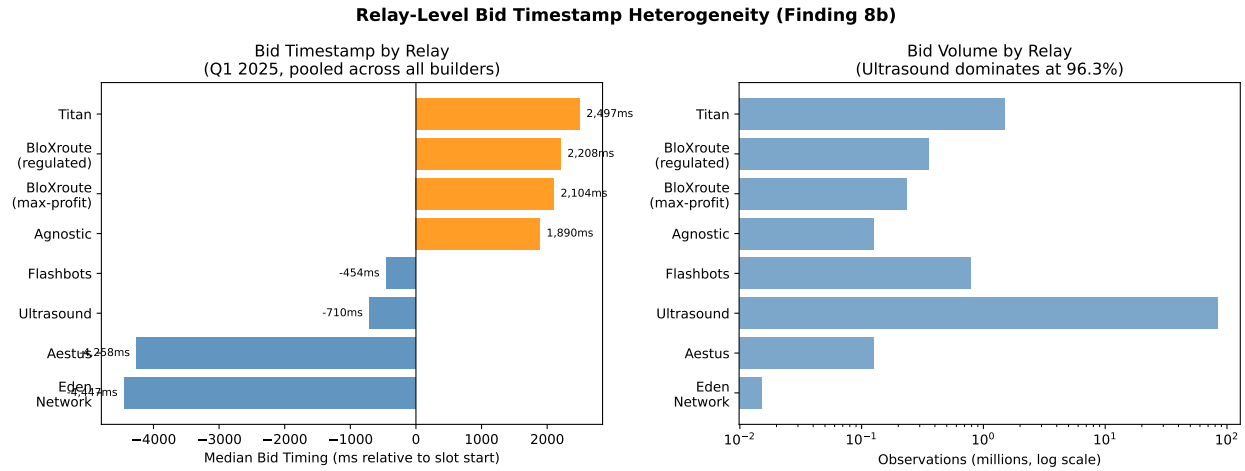



Figure 15: Relay-level bid timestamp heterogeneity, full Q1 2025 bid archive (86.8M observations). Left: median bid timing by relay, pooled across all builders; positive values indicate post-slot-time reporting. Right: bid volume by relay (log scale); Ultrasound's websocket stream dominates at 96.3% of all observations and reports systematically earlier timestamps than other relays.

Table 11 reports GARCH(1,1) parameter estimates on the full 1,219,851 bid margin observations. As with the block value series (Finding 2), parameter estimates show sensitivity to optimizer starting values; we report the result obtained under default initialization, cross-validated against a winsorized (1%) specification that converges to an estimate within 0.0002 of the primary result.

Table 11: GARCH(1,1) on bid margins, full sample

| Specification | $\hat{\alpha}+\hat{\beta}$ | $\hat{\nu}$ | AIC |
|---|---|---|---|
| Default initialization | 0.9796 | 4.230 | 15,496,738 |
| Winsorized (1%) | 0.9798 | 4.298 | — |

$n = 1{,}219{,}851$. Both specifications agree to within 0.0002, supporting 0.980 as the robust estimate.

Bid margin volatility persistence ($\hat{\alpha}+\hat{\beta} \approx 0.980$) is statistically indistinguishable from block value volatility persistence ($\hat{\alpha}+\hat{\beta} = 0.980$, Finding 2). Bid margin dynamics therefore inherit the same near-integrated volatility persistence as the underlying block value process, with a similarly heavy-tailed error distribution ($\hat{\nu} \approx 4.2$–$4.3$), indicating that competitive bidding pressure does not measurably dampen MEV volatility shocks at the auction-margin level.

> **Finding 9.** *Bid margin volatility is highly persistent ($\hat{\alpha}+\hat{\beta} \approx 0.980$) with heavy tails ($\hat{\nu} \approx 4.2$–$4.3$), statistically indistinguishable from block value volatility persistence (Finding 2). Competitive bidding pressure does not measurably dampen MEV volatility shocks at the auction-margin level; both the underlying MEV opportunity and the competitive response to it exhibit the same near-integrated volatility dynamics.*

### 5.6.7 Concentration-Competitiveness Decoupling

A natural hypothesis is that higher builder concentration (HHI) predicts lower auction competitiveness (higher bid margins). We test this directly by regressing weekly median bid margins on lagged weekly HHI across 89 weeks.

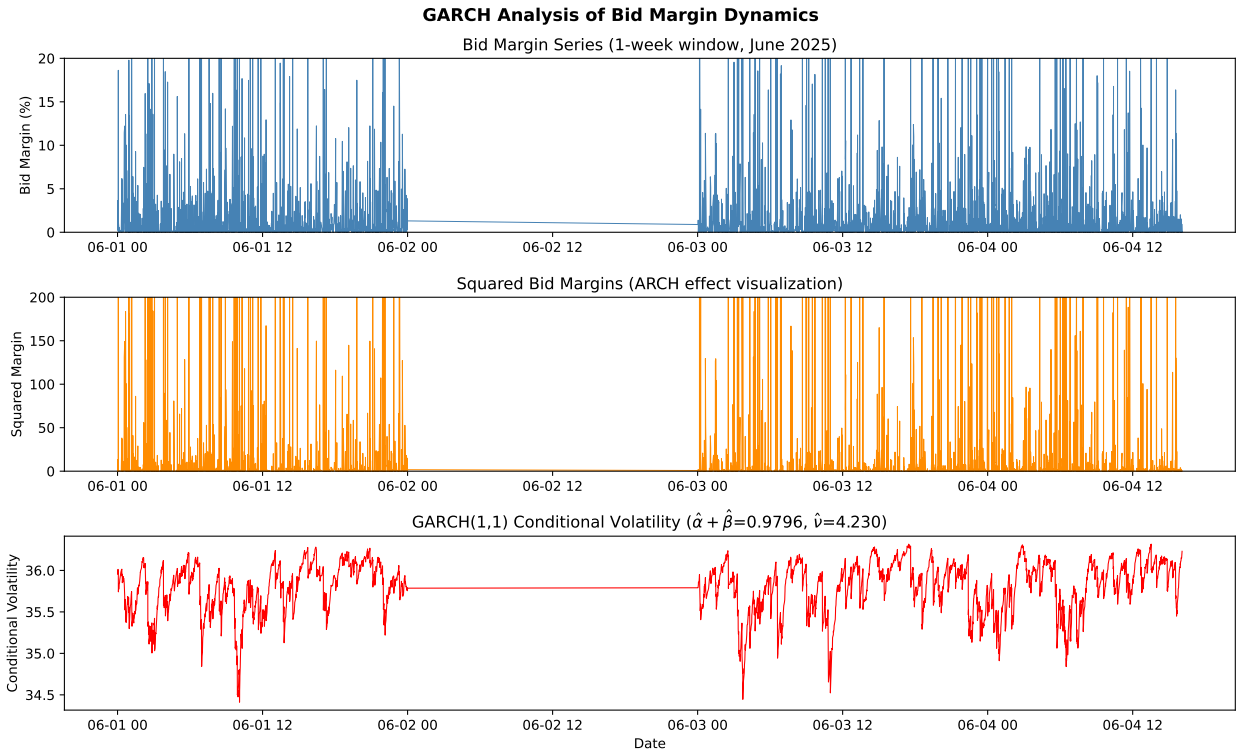



Figure 16: Top: bid margin series (one-week window, June 2025). Middle: squared bid margins illustrating ARCH-type clustering. Bottom: GARCH(1,1) conditional volatility ($\hat{\alpha} + \hat{\beta} \approx 0.980$, $\hat{\nu} \approx 4.2$).

The result is a null: lagged HHI does not significantly predict bid margin (Spearman $\rho = 0.075$, $p = 0.486$; OLS $R^2 = 0.005$). Builder concentration at the delivery level (who wins blocks) is decoupled from auction competitiveness at the bid level (how hard builders compete). This finding has an important implication: even when a small number of builders dominate block delivery, the bidding process remains competitive because the same builders still face active competition for each individual slot. Concentration in outcomes does not imply concentration in competition.

This result reinforces the PBS design rationale: the separation of block construction from block proposal creates competitive pressure at the auction level that persists even as market structure evolves at the builder level.

> **Finding 10 (null result).** *Builder concentration (weekly HHI) does not predict auction competitiveness (weekly median bid margin; $\rho = 0.075$, $p = 0.486$). Concentration in block delivery outcomes is decoupled from competition in the bidding process, consistent with the PBS design goal of maintaining competitive pressure independently of market structure.*

### 5.6.8 Proposer Revenue Efficiency

We define proposer revenue efficiency as $1 - m_s$, the fraction of the winning bid that the proposer receives (where $m_s$ is the bid margin from eq. 9). Across 1,219,851 matched slots, median proposer revenue efficiency exceeds 99.7% in all value terciles (Table 12), confirming that proposers capture nearly the full winning bid value in typical slots.

Table 12: Proposer revenue efficiency by block value tercile

| Value Tercile | Mean Efficiency (%) | Median Efficiency (%) |
|---|---|---|
| Low | 93.16 | 99.78 |
| Mid | 95.74 | 99.85 |
| High | 97.15 | 99.89 |

The median efficiency is highest in the high-value tercile (99.89%), the precise regime where MEV opportunity is greatest. The mean-median gap in the low-value tercile (93.16% mean vs 99.78% median) reflects the right tail of the bid margin distribution documented in Section 5.5.1—a small fraction of low-value slots exhibit near-monopoly conditions in which builders retain substantial surplus.

An intraday pattern in proposer efficiency is also apparent: efficiency is lowest during hours 01:00–03:00 UTC (corresponding to the Asian overnight session, when builder participation is thinnest) and highest during 13:00–22:00 UTC (US daytime and evening). This pattern mirrors the temporal seasonality in block values documented in Finding 4.

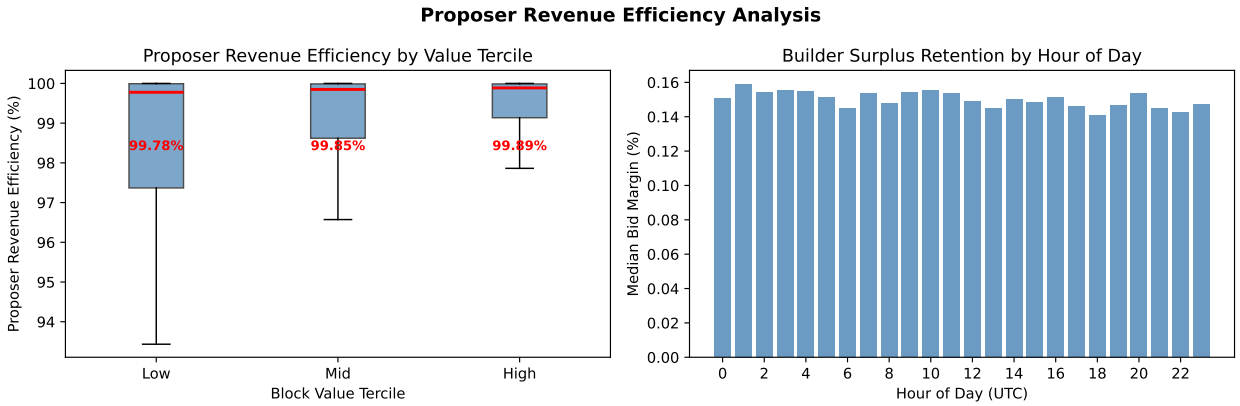



Figure 17: Left: proposer revenue efficiency by block value tercile (median annotated). Right: median builder surplus retention (bid margin) by hour of day (UTC); builder surplus is highest during Asian overnight hours when participation is thinnest.

**Finding 11.** *Proposers capture a median of 99.78–99.89% of the winning bid value across all block value terciles, with efficiency highest in high-value slots. Builder surplus retention is greatest during Asian overnight hours (01:00–03:00 UTC) when participation is thinnest, mirroring the intraday seasonality pattern in block values.*

# 6 Discussion

## 6.1 Implications of High Volatility Persistence

The GARCH(1,1) estimate of $\hat{\alpha} + \hat{\beta} = 0.980$ on the full 20-month sample, indicates that MEV volatility is highly persistent. In financial econometrics, persistence values above 0.95 are considered extreme; the MEV estimate of 0.980 implies that a volatility shock today retains approximately

82% of its impact after ten periods and 13% after one hundred periods. For PBS participants, this means that elevated MEV periods are not quickly transitory but persist for extended durations.

The full-sample persistence of 0.980, confirmed across specifications, establishes that MEV volatility clustering is highly persistent and that shocks propagate for extended durations. Subperiod estimation is not reliable for this series due to the mixture-like distribution of $\log(1 + v_t)$ returns at sub-annual horizons: rare extreme blocks drive non-stationarity within individual subperiods, causing optimizer non-convergence. The full-sample result should therefore be interpreted as characterising the long-run volatility dynamics of the PBS market as a whole, rather than implying a specific trajectory of persistence over time.

### 6.2 Cyclical Concentration Dynamics

The 20-month HHI series reveals cyclical concentration dynamics not observable in shorter samples. The secondary concentration episode in mid-2025, where HHI briefly returned to approximately 1,500, suggests that builder market structure responds to external shocks, likely periods of high MEV opportunity that favor builders with superior infrastructure or exclusive orderflow access [Wu et al., 2025]. This has implications for censorship resistance analysis: even in an ostensibly competitive market, concentration can temporarily reemerge during high-value periods.

### 6.3 Cross-Regime Robustness

The core findings of Sections 5.1–5.5 hold across distinct market regimes in the extended sample. The September–December 2024 period, the 2025 market cycle, and the January–May 2026 period represent meaningfully different conditions. The consistency of patterns across these regimes supports the interpretation that the findings reflect structural features of PBS rather than artifacts of a specific market environment.

### 6.4 Auction Efficiency and Proposer Revenue

The auction microstructure findings in Section 5.5 have direct implications for the efficiency of the PBS mechanism. A median bid margin of 0.141% implies that, in typical slots, proposers receive approximately 99.86% of the winning builder's top bid as their payment. The compression of margins in high-value slots (to 0.110% at the median for the top tercile) means that auction efficiency is greatest precisely when MEV opportunity is largest—a desirable property from the perspective of validator economics.

However, the right tail of the margin distribution (90th percentile 6.691%) indicates that a non-trivial fraction of slots exhibits materially reduced competition, in which builders retain a substantially larger fraction of block value. Section 5.6.7 tests directly whether temporal variation in bid margins is explained by the weekly HHI series and finds no such link, indicating that these two dimensions of market structure move independently.

The decline in builder participation (from approximately 17 to 9–11 builders per slot) also contextualises the HHI deconcentration. The HHI measures concentration in delivered blocks (the winning side), while builder participation measures the full set of competitors. That the HHI fell while the number of active bidders also fell indicates that deconcentration was achieved through exit of marginal builders and a more even split among the survivors, not through entry. This is consistent with a market in which the fixed costs of competitive block building (orderflow access, latency infrastructure) select for a small set of comparably capable builders.

### 6.5 Limitations

The volatility measure is constructed from block values themselves rather than external ETH price data, limiting the ability to separate market-driven from MEV-specific volatility components. Alternative asymmetric GARCH specifications (EGARCH) did not yield stable estimates on this dataset and are not interpreted. The non-standard mixture-like distribution of $\log(1 + v_t)$ returns means that subperiod GARCH estimation is not reliable for this series, and alternative volatility models (e.g. stochastic volatility, EGARCH) may better capture the data-generating process at sub-annual horizons.

BloXroute contributed approximately 79% of the deduplicated sample; if BloXroute's relay coverage is systematically different from the broader relay market, results may not fully generalize. This concentration is a direct consequence of the data collection process: relay selection was constrained to publicly-queryable endpoints, and within that set, API availability rather than market share determined each relay's representation. A follow-up collection effort using authenticated access across a broader and more balanced set of relays would strengthen the external validity of the findings.

Bid-level data from relay APIs was unavailable for direct collection due to authentication restrictions. The auction analysis in Section 5.5 instead uses the Relayscan.io Bid Archive [Relayscan, 2024]. A limitation of the archive data is that the `slot_t_ms` timing field reflects when bids were received by the Relayscan collector rather than necessarily when they were submitted by the builder; absolute timing should therefore be interpreted with caution, though relative comparisons across slots and value terciles remain valid. Additionally, the "top bid" files record only the highest bid per builder per slot, so intra-builder bid revision behaviour cannot be characterised from these data.

Builder concentration measures are computed at the pubkey level rather than the entity level. Entity-level relabeling was attempted by examining the `proposer_fee_recipient` field, but this field identifies the validator's payout address rather than the builder's identity (32,376 unique values, consistent with the size of the active validator set), and is therefore unsuitable for builder entity resolution. Reliable entity-level labeling would require either an `extra_data` field, which was not captured during data collection, or an external pubkey-to-entity mapping, which is not consistently available in public form across all six relays. If multiple pubkeys are operated by a single entity, reported HHI and Gini values understate true concentration; the 2,100→600–1,000 HHI decline and 0.85→0.77 Gini decline should be interpreted as upper bounds on the rate of genuine decentralization.

## 7 Conclusion

This paper characterises MEV block values as a stochastic process across multiple market regimes, using 1,271,079 delivered payload observations from six MEV-Boost relays spanning September 2024 through May 2026, supplemented by 1,220,100 slot-level bid observations from the Relayscan.io Bid Archive. Eleven findings emerge, all estimated on the full 20-month sample:

1. **Volatility regime sensitivity:** Mean block values are approximately 7.0× higher in high-volatility regimes with variance approximately 98× larger ($\rho = 0.494$, $H = 266{,}905$, $p < 10^{-300}$).

2. **Highly persistent volatility dynamics:** Block value log returns exhibit high volatility persistence ($\hat{\alpha} + \hat{\beta} = 0.980$) with very heavy tails ($\hat{\nu} = 4.005$), confirmed across GARCH(1,1)

and winsorised specifications. A persistence of 0.980 implies that volatility shocks retain approximately 82% of their impact after ten periods. Subperiod GARCH estimation is not reliable for this series due to the non-standard mixture-like distribution of $\log(1+v_t)$ returns.

3. **Cyclical deconcentration:** Builder HHI declined from approximately 2,100 in September 2024 to a stable range of 600–1,000, with a secondary concentration episode in mid-2025 (HHI $\approx$ 1,500). The Gini declined from 0.85 to a long-run mean of 0.77.

4. **Robust temporal seasonality:** Both intraday ($H = 65{,}875$) and day-of-week ($H = 33{,}590$) patterns are highly significant across the full 20-month period. Median block values peak during 13:00–17:00 UTC and are highest on Mondays.

5. **Competitive but consolidating auction:** Across 1,220,100 matched slots, the PBS auction exhibits high competitiveness (median 132 bids, 11 builders, bid margin 0.141%). Bid margins compress monotonically with block value (0.196% to 0.110% across terciles; $\rho = -0.054$, $p < 10^{-300}$), indicating that high-value slots are allocated most efficiently. Builder participation declined from approximately 17 to 9–11 per slot over the sample while bid counts and margins fluctuated without trend, indicating consolidation among active builders rather than deepening.

6. **MEV is orthogonal to ETH price:** ETH price explains essentially zero variance in block value in ETH terms ($R^2 = 0.0001$, $\rho = 0.050$). MEV block value is driven by on-chain activity, not ETH price movements.

7. **Relay participation deepens competition:** Bid margins decline monotonically from 0.217% in single-relay slots to 0.066% in 7-relay slots ($\rho = -0.057$, $p < 10^{-300}$). Relay participation is a structural determinant of auction depth.

8. **Builder strategy heterogeneity:** Two distinct builder strategies are empirically identifiable: late-aggressive high-value bidders and early-conservative high-frequency bidders, consistent with differential access to exclusive orderflow.

9. **Margin volatility matches block value volatility:** Bid margin persistence ($\hat{\alpha} + \hat{\beta} \approx 0.980$) is statistically indistinguishable from block value persistence (0.980), indicating that competitive bidding pressure does not measurably dampen MEV volatility shocks at the auction-margin level.

10. **Concentration-competitiveness decoupling (null result):** Lagged HHI does not predict bid margins ($\rho = 0.075$, $p = 0.486$). Builder concentration in delivery outcomes is decoupled from slot-level auction competitiveness.

11. **Near-perfect proposer revenue efficiency:** Proposers capture a median of 99.78–99.89% of the winning bid value across all value terciles, with efficiency highest in high-value slots and lowest during Asian overnight hours.

Future work should characterise intra-builder bid revision behaviour using full bid-stream data; formally test whether bid margin dynamics are driven by MEV-specific versus ETH-price-driven volatility components; apply regime-switching models to characterise structural breaks in volatility persistence; extend the concentration analysis to entity-level builder labels; and pursue authenticated relay access across a broader, market-share-weighted relay set to assess the sensitivity of the reported findings to the BloXroute-dominated composition of the current sample.

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